DES-345 Image options in multiple choice assessments

PRD and design brief · 20 Oct 2026 · Reporter and specs owner: Nicolas Darriulat · Design: Bruno Carvalho · Customer: Park Stars Entertainment (critical) · Planned release: 27 Nov 2026

1. Context

The problem

Park Stars Entertainment trains frontline staff on things that are easier to recognise than to describe: the right wristband, the correct safety sign, a clean versus a dirty station. They need multiple choice questions where the answers are pictures, and several can be right. **Table stakes:** Kahoot, Google Forms, Interact and LifterLMS all ship image answers, and multi-select via checkboxes is standard. **Novel for 5Mins:** image-only options with no text label, a multi-correct question in the course builder (today it allows one correct answer), and a classic assessment that learners can actually play on web and mobile, which the prototype does not yet do.

What the code does today

Area	Today	Gap for DES-345
Course builder MCQ editor	`AssessmentModal.tsx` (`src/pages/your-courses/components/AssessmentModal/`), type `'single-choice'` , title "Add assessment - Multiple Choice", subtitle "Users can select one correct choice". Options are `string[]` ; one `correctIndex: number` set with DS `Radio` plus a Success `Badge` "Correct". One question-level image or audio with crop. Footer `Save` only. Opened inside `ContentDrawer.tsx` .	No option mode, no image per option, no multiple correct, no Preview.
Preview pattern	`InteractiveDrawer.tsx` : `Save` + outlined-2 `Preview` ; opens `ConfirmModal + DesktopPreview` (544x812, the learner panel width) running the real learner renderer with `QuizHeader` .	Reuse as is for the MCQ.
Question bank editor	`QuestionEditor.tsx` with `mockQuestions.ts` : `multiple_choice` and `multi_select` ("Select all that apply"), answers `{ id, text, isCorrect }` , 2 to 6. Not built from DS controls.	Out of scope (D5); stays text only.

Learner renderers	`src/pages/quiz-lab/formats/SingleChoice.tsx` and `SelectAll.tsx`, `FeedbackFooter` (`Check`, `Continue`), `ResultBanner` ("Well done!", "Not quite!"), `quiz-lab.css` (`--selected` fill, success and danger tints). Untouched options are never graded.	`SelectAll` reveals the count ("Select Four", "Select 2 More", caps at 4) and turns a wrong set all red. Both conflict with decisions 2 and 4. No image tile.
Learner web	`ProgramCourseDetails.tsx` (`/courses/:id`) plays authored interactive questions in a `pcd-quizstage` dialog; `previewCourse.ts` only carries `InteractiveQuestion`s.	Classic MCQ is not playable on learner web.
Learner mobile app	`CourseScreen.tsx` lists `MobileAssessmentCard`s with no `onClick`. App is dark only via `PhoneFrame`.	No assessment-taking screen.
Admin results	`AssessmentAnswers.tsx`, `AnswerStats.tsx`, `assessmentResults.ts` key everything on option text and one `correctIndex`.	Image options would show blank labels; D7 makes them show thumbnails.
DS parts	`AddImageModal` (upload, Generate With AI, Stock Footage; used for course, lesson and program thumbnails), which wraps `FileUploader` (Enabled, Uploading with `progress`, Error with `errorMessage`, Filled with `filledPreview`), `Checkbox`, `Radio`, `ConfirmModal`, `Tooltip`, `Toast`.	No documented image option tile, admin or learner. Figma `Quiz/Options` (`5504:24966`) is text only and has no code component.

Decisions taken

#	Topic	Decision
D1	Assessment type	Image Choice is its own assessment type, listed after Multiple Choice in Add Content > Assessments (changed 20 Oct; was a Text/Image switch inside Multiple Choice). Every option is an image; text and image options never mix.
D2	Scoring	All-or-nothing: correct only if the learner selects exactly the correct set. The prompt says "Select all that apply" and never reveals how many are correct; after Check the feedback gives counts (20 Oct, D12).
D3	Alt text	No alt text field. Screen readers hear "Option 1", "Option 2" and so on. WCAG 1.1.1 is logged as a risk, not a blocker.
D4	Feedback after submit	Every image stays after Check (changed 20 Oct; was picks only). Right picks: green with a tick. Wrong picks: red with a cross. Correct images not picked: dashed green frame, no badge. Unpicked wrong images: unmarked. Prior art: Canvas, Quizlet, Open edX.

D5	Where it is built	Create Course > Assessments only (`AssessmentModal` in the course builder). The question bank and lesson quizzes do not get image answers.
D6	Correct answers per type	Multiple Choice stays single-correct. Several correct answers exist only in Image Choice.
D7	How image options are shown to admins	Image thumbnails, never "Option N", in both the assessment preview and the admin answers view (`AssessmentAnswers` , `AnswerStats`): wherever admins see learner answers to an image question.
D8	File types	JPG, PNG, WebP and GIF.
D9	Option count	The form starts with 2 image options; admins can add as many as they want. No maximum.
D10	Learner button	Stays "Check", then "Continue", following the same patterns as the other assessments.
D11	Shuffle	Not shuffled. Image options appear in the order the admin set them, on every attempt.
D12	Partly correct answer	The marks on the images explain the result (D4), so the banner shows only a heading: "Partly right" when every pick was right but some were missed, "Not quite!" otherwise, followed by the admin explanation if there is one. Screen readers hear the counts: "You got {n} of {total} correct images, picked {w} wrong and missed {m}." (20 Oct).
D13	Image source	Same pattern as course and lesson thumbnails: each option opens the existing `AddImageModal` (`src/pages/add-content/components/AddImageModal/AddImageModal.tsx`) with its three default tabs, "Upload image", "Generate With AI" and "Stock Footage".

Acceptance criteria

The ticket has no formal ACs; the "What to design" bullets and states are quoted verbatim and numbered.

1. MCQ editor (admin web): "set each option as an image"
2. MCQ editor (admin web): "upload and replace an image"
3. MCQ editor (admin web): "mark one or more options as correct"
4. MCQ editor (admin web): "preview the question as learners will see it"
5. MCQ editor states: "empty option, uploading, uploaded, upload error"
6. MCQ question with image options (learner web and learner mobile app): "view the image options"
7. Learner web and mobile: "select one or more answers and submit"
8. Learner states: "unselected, selected, correct, incorrect"

9. Deliverables: "Admin web, learner web and the learner mobile app"

Out of scope

Image answers in the question bank (`QuestionEditor`) and lesson quizzes; image options in polls and AI generation; multiple correct answers in text MCQs; text labels on image options; mixing text and image options; partial credit; an alt text field; revealing missed correct answers; video or audio as options; redesigning the admin answers view beyond showing thumbnails.

2. Research

Full notes, including Mobbin screenshots, are in `2026-10-20-DES-345-image-options-mcq.research.md` .

Authoring image options

Source	What it does	Adopt or reject
Interact image answers	One question-level "Use Image Answers" switch; recommends 300x250px.	Rejected (D1): a separate Image Choice type replaces the switch.
Kahoot images as answers	Image button beside each answer; PNG, JPEG, GIF; up to 80MB and 3264x3264px.	Adopt per-option upload. Reject 80MB; too heavy for a mobile feed.
TalentLMS question types	Multiple response with checkboxes; media only above answers; 10MB image cap.	Adopt checkboxes for "correct" and a 10MB cap.
Typeform image sizes and community	Resizes uploads (800px, 460px mobile); learners find picture tiles too small.	Adopt server resize; keep tiles large (2 columns).
LifterLMS picture multiple choice, multiple correct	Dedicated type for picture options with several correct.	Confirms table stakes.
Google Forms image answers	Image per option on Multiple choice and Checkboxes, always with a text label.	Adopt checkbox semantics for multi-select; reject labels for now.
Sana question generator	MCQ generation documented.	Sana yielded nothing on image answer options.
Docebo community, Cornerstone release note	Images in question stems only. Workday Learning: nothing found.	Nothing usable.

Selecting, grading and accessibility

Source	What it does	Adopt or reject
Duolingo exercise types	Pick-the-picture grid, Check, then green or red.	Adopt; quiz-lab already follows this rhythm.
UGA multi-select grading	All-or-nothing versus right-minus-wrong.	All-or-nothing (D2).
NYU WCAG 1.1.1 notes	An image that is a control's only label fails 1.1.1 and 4.1.2 without a text alternative.	Logged as a risk (D3).
VCCS accessible assessment	Alt text must describe what is visible without giving the answer away.	Input for a later alt text phase.

Mobbin

Screen	What it shows	Adopt or reject
MacroFactor body fat picker	Image grid; selected tile gets a heavy border plus a tick.	Adopt: selection never by colour alone.
Shopify select files	Corner checkbox on each photo, "3 selected" count.	Adopt the corner checkbox. Reject the count; it hints at the answer count.
Lloyds choose an image	2-column tiles, CTA pinned at the bottom.	Adopt 2 columns and the pinned CTA (already `FeedbackFooter`).
Mintlify select image	Upload tile sits inside the grid of images.	Adopt: an empty option is an upload tile in the grid.
Recraft upload	Formats line next to the upload.	Adopt the formats hint in the field label.
Bloom logo issue	Error in place, with reason and re-upload.	Adopt (FileUploader Error state).
fal image inputs	Error shown away from the failed slot.	Reject.

Mobbin had no admin editor where every answer is an image upload; the editor is assembled from the upload-tile and image-grid patterns above.

Build For Mars

Lesson	Principle	Adopt or reject
Hinge's photo tips	A grid of upload tiles plus a short tip on what works.	Adopt a one-line hint.

Incorrect recipient details (Monzo)	Mark each item right or wrong in place.	Adopt per-pick marks (D4).
Creating confidence from errors (Lovable)	An error says what happens next.	Adopt: every upload error names the fix.
Animated quiz master (Ahead)	Small motion makes answering feel responsive.	Adopt the existing quiz-lab select cue.

BFM had no lesson specific to image-only multi-select quizzes.

3. Implementation

Blockers

#	Blocker	Recommendation
B1	**Makes AC3 unbuildable.** `AssessmentData.correctIndex` is one number, and the editor uses a `Radio`. Several correct answers cannot be stored.	Change options to `{ id, text?, image?, isCorrect }[]` (the shape `mockQuestions.ts` already uses), with `optionMode: 'text' | 'image'` on the question. Text mode is unchanged: `Radio`, exactly one correct, and the existing "Users can select one correct choice" subtitle (D6). Only image mode uses `Checkbox` and allows one or more correct; switching back to text never carries several correct flags over.
B2	**Makes AC1, AC5, AC6 and AC8 unbuildable to DS standard.** No Figma or doc for the admin image option tile (upload, correct, replace, remove) or the learner selectable image tile (unselected, selected, correct, incorrect, focus, both themes).	Ask for Figma for both before build. Proposed admin tile: the course thumbnail box (`cdt-thumb__box`, empty and filled) that opens `AddImageModal` (D13), a `Checkbox` "Correct" under it, and icon-only Replace and Remove with `Tooltip`. Proposed learner tile: 1:1 image in a `ql-token` -style frame, corner `Checkbox` glyph on select, `--selected` border, success and danger tints plus `TickCircle` and `CloseCircle` icons after Check.
B3	**Makes AC6 to AC8 unbuildable on learner surfaces.** Classic MCQs have no learner screen: web only plays interactive formats, mobile cards have no tap.	New renderer `ImageChoice` in `src/pages/quiz-lab/formats/`, hosted in the existing `pcd-quizstage` (web) and a new branch from `MobileAssessmentCard` `onClick` in `CourseScreen.tsx` (mobile, inside `PhoneFrame`).

B4	Image rules: file types and option count are now set (D8, D9); size cap and aspect ratio still need Nicolas's sign-off.	JPG, PNG, WebP and GIF; max 10MB each; square 1:1 crop using `AddImageModal` 's `upload.confirm` zoom-and-Save step (D13). That step re-encodes to PNG through a canvas, which would freeze a GIF, so GIFs skip it and keep their animation, framed square by the tile instead. 2 options by default, no maximum.
B5	Correct picks can sit beside a wrong result: with D2 and D4, a learner who picks 2 of 3 correct images sees 2 green marks but fails. The ticket does not define this state.	**Resolved (20 Oct, D4 and D12).** Missed correct images are shown with a dashed green frame; the banner heading says "Partly right" or "Not quite!".

Proposed behaviour and copy

Copy follows the 5mins-copy-review skill; button copy is Title Case per house rules.

- **Entry point (AC1).** "Image Choice" in Add Content > Assessments, after Multiple Choice, with the Iconsax `Gallery` icon. Form title "Add assessment - Image Choice", subtitle "Users can select one or more correct images". No mode switch: Multiple Choice keeps its text form unchanged. The outline card borrows the Multiple Choice illustration until design supplies one.
- **Options grid (AC1, AC2, AC5).** Starts with 2 empty tiles (D9). Label "Images (JPG, PNG, WebP, GIF • max. 10MB)" with hint "Use clear, similar-sized photos. Learners see them as squares." Empty tile: dashed box with `GalleryAdd` and an "Add Image" button, as in the course thumbnail row (`CourseDetailsTab.tsx`). Clicking it opens `AddImageModal` (D13) with the default tabs `['upload', 'generate', 'freepik']` , `showSuggest={false}` , and `upload` set to `accept: '.jpg,.jpeg,.png,.webp,.gif'` , a `validate` for the 10MB cap and `confirm: true` for the square crop (skipped for GIFs, B4). Uploading, AI generating and stock loading show the modal's own progress. Uploaded: the tile shows the image, then icon buttons "Replace image" (reopens `AddImageModal`) and "Remove option" (aria-label and Tooltip). Upload errors show inside the modal's `FileUploader` : "This image is over 10MB. Choose a smaller file.", "Use a JPG, PNG, WebP or GIF image.", "Upload failed. Try again." The existing "Add Option" button adds a tile, with no limit.
- **Correct answers (AC3).** `Checkbox` labelled "Correct" under each uploaded tile, aria-label "Mark option 2 as correct". Save stays disabled until there are at least 2 uploaded images and 1 correct, and no upload is in progress.
- **Preview (AC4).** Outlined-2 "Preview" beside "Save", as in `InteractiveDrawer` , opening `ConfirmModal` + `DesktopPreview` with the learner renderer, so the admin sees the image thumbnails, not "Option N" (D7).
- **Learner (AC6 to AC8).** Question text, then "Select all that apply" in `--text-secondary` . Options are laid out in a 2-column grid of square tiles in both the 544px web panel and the 375px

phone. Each tile is a checkbox-role button named "Option 1", "Option 2" and so on, in the order the admin set them (D11). `FeedbackFooter` `Check` (D10) is disabled until 1 pick and never shows a count. After Check: picks marked correct or incorrect; unpicked tiles dim and stay unmarked; banner "Well done!" or "Not quite!" (plus "You missed at least one correct image." when a correct image was missed, D12); then `Continue`.

- **Themes.** Admin and learner web in light and dark; the mobile app is dark only. Semantic tokens only (`--selected`, `--text-on-selected`, `--border-elevated`, success and danger tints as in `quiz-lab.css`).

Phases

#	Change	ACs	What verifies it
0	Figma for both tiles (B2); Nicolas confirms the size cap and crop (B4) and B5.	1, 5, 6, 8	Figma links in the ticket.
1	**Done (20 Oct).** Data model: a new `image-choice` assessment type with `optionImages` and `correctIndices`; Multiple Choice limited to one correct; all-or-nothing scoring helper keyed on option ids.	3, 7	Vitest: exact set passes; subset, superset and mixed fail; text mode rejects a second correct. `pnpm build`.
2	**Done (20 Oct).** Editor: Image Choice menu entry and form (replaced the Option type radio on 20 Oct), image grid of thumbnail tiles opening `AddImageModal` (upload, generate, stock), Checkbox correct, replace and remove, validation.	1, 2, 3, 5	/test-feature on the course builder: each state reachable, keyboard only, both themes.
3	**Done (20 Oct).** `ImageChoice` renderer, with a `/quiz-lab` entry for review.	6, 7, 8	Vitest for grading and for "no count shown"; screen reader names "Option N" follow the admin's order.
3a	~~Shuffle per attempt~~ Dropped (D11): options keep the admin's order.	n/a	n/a
4	**Done (20 Oct).** Editor Preview button.	4	Preview renders the same component as phase 3 at 544px.
5	**Done in Quiz Lab (20 Oct):** the image question plays at /quiz-lab, Mobile and Desktop, instead of the course page or mobile app. Original plan: Learner web: `pcd-quizstage` branch and `previewCourse.ts` carries the MCQ.	6, 7, 8, 9	Play it from builder Preview at `/courses/:id`.

6	**Done in Quiz Lab (20 Oct):** the image question plays at /quiz-lab, Mobile and Desktop, instead of the course page or mobile app. Original plan: Mobile: `MobileAssessmentCard onClick` in `CourseScreen.tsx` opens the renderer in `PhoneFrame`.	6, 7, 8, 9	/test-feature on `/mobile`, dark theme.
6a	**Done (20 Oct).** Admin answers view: `AnswerStats` and `AssessmentAnswers` show image thumbnails for image options (D7).	9	An image MCQ's results show thumbnails, not blank labels or "Option N".
7	Hand-off with /figma-ship.	9	Screens in the agreed Figma section.

Risks to watch during build

- WCAG 1.1.1 and 4.1.2: "Option N" is a name, not a text alternative, so a screen reader user cannot answer. Keep this visible to Park Stars (D3).
- Reusing `SelectAll` as is would reveal the answer count and turn all picks red; fork the grading, do not reuse its CTA logic.
- `.ql-` and `.assessment-modal-` classes are globally bundled; grep before adding tile classes.
- No option maximum (D9): a long list makes a tall grid on a 375px phone, and many large images or animated GIFs slow the feed. Mitigation, not a cap: serve resized images (Typeform pattern), lazy-load tiles below the fold, play GIFs only while on screen and respect reduced motion, keep the `Check` footer pinned so it stays reachable.
- `AddImageModal`'s square-crop step (`confirm`) outputs a PNG, so an uploaded GIF would lose its animation. Branch on file type so GIFs bypass it (B4), and check this in phase 2.
- `AddImageModal` currently returns a data URL. Fine for the prototype, but many options per question means production needs real upload storage and resizing (see the option-count risk above).
- `AnswerStats` and `AssessmentAnswers` will show empty labels for image options until phase 6a lands.

Still open

Nicolas's sign-off on the 10MB cap and square crop (B4).